THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 8 (2020/2021)

1. For each of the following functions, determine whether the limit exists or does not exist at a given value of x.

(a)
$$f(x) := \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x > 0, \end{cases} \quad \text{at } x = 0$$

(b)
$$f(x) := \begin{cases} 7x - 2 & \text{if } x \geq 2, \\ 3x + 5 & \text{if } x < 2, \end{cases} \quad \text{at } x = 2$$

(c)
$$f(x) := \frac{x^2 - 16}{x - 4}, \quad x \neq 4, \quad \text{at } x = 4$$

(d)
$$f(x) := \begin{cases} x^2 + 5 & \text{if } x < -2, \\ 3 - 3x & \text{if } x \geq -2, \end{cases} \quad \text{at } x = -2$$

2. Evaluate each of the following:

(a) i.
$$\lim_{x\to 0} \left(\frac{1}{x} - \frac{1}{x^2}\right).$$

ii.
$$\lim_{x\to -5} \frac{x^2}{3x - 4}$$

iii.
$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3}$$

iv.
$$\lim_{x\to 2} \frac{x^3 - 8}{x - 2}$$

v.
$$\lim_{x\to 2} \sqrt{\frac{x^2 - 3x + 2}{x^2 - 4}}$$

vi.
$$\lim_{x\to 1} \frac{x^3 + x^2 - x - 1}{x - 1}$$

vii.
$$\lim_{x\to 1} \frac{x^4 + 3x^3 - 13x^2 - 27x + 36}{x^2 + 3x - 4}$$

viii.
$$\lim_{x\to 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$$

(b) Evaluate each of the following:

i.
$$\lim_{x\to +\infty} 2x^{11} - 5x^6 + 3x^2 + 1$$

ii.
$$\lim_{x\to -\infty} 3x^4 - x^2 + x - 7$$

iii.
$$\lim_{x\to +\infty} \frac{2x + 5}{x^2 - 7x + 3}$$

iv.
$$\lim_{x\to +\infty} \frac{3x^3 - 4x + 2}{7x^3 + 5}$$

v.
$$\lim_{x\to +\infty} \frac{4x^5 - 1}{3x^3 + 7}$$

vi.
$$\lim_{x\to +\infty} \frac{4x - 1}{\sqrt{x^2 + 2}}.$$

vii.
$$\lim_{x\to +\infty} \frac{\sqrt{x^2 + 5}}{3x^2 - 2}.$$

3. (a) Discuss the continuity of each of the following functions at a given value of x.

i.
$$f(x) := \begin{cases} x^2 & \text{if } x \leq 0, \\ x & \text{if } x > 0, \end{cases} \quad \text{at } x = 0$$

ii.
$$f(x) := \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x > 0, \end{cases} \quad \text{at } x = 0$$

iii.

$$f(x) := \begin{cases} \frac{x^2-4}{x+2} \text{ if } x \neq -2 \end{cases} \quad \text{at } x = -2$$

iv.

$$f(x) := \begin{cases} x+1 & \text{if } x \geq 2, \\ 2x-1 & \text{if } 1 < x < 2, \\ x-1 & \text{if } x \leq 1, \end{cases} \quad \text{at } x = 1 \text{ and } x = 2$$

4. (a) Find the derivative of each of the following from the first principles:

i.

$$y = 5$$

ii.

$$y = 2x$$

iii.

$$y = 2x^2 + 3x + 6$$

iv.

$$y = \frac{1}{x+2}$$

v.

$$y = \sqrt{x+1}$$

vi.

$$y = \frac{1}{\sqrt{x+1}}$$

(b) Differentiate each of the following with respect to x :

i.

$$y = x + \frac{1}{x}$$

ii.

$$y = x^2 - \frac{8}{\sqrt{x}}$$

iii.

$$y = (x+2)^2$$

iv.

$$y = \frac{3x^7 + 5x^5 - 2x^4 + x - 3}{x^4}$$

v.

$$y = 2x^3 + \sqrt{x} + \frac{x^2 + 2x}{x^2}$$

(c) i. Show that

$$f'(x) = \frac{3}{2}x^{-\frac{1}{2}}(4-x),$$

if

$$f(x) = 12x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

ii. Expand the right hand side of

$$f(x) = (x^{\frac{3}{2}} - 1)(x^{-\frac{1}{2}} + 1),$$

and hence find $f'(x)$.

(d) Difference with respect to x :

i.

$$f(x) = (3x^4 - x^2)^5$$

ii.

$$f(x) = \sqrt{x-2}$$

iii.

$$f(x) = x^2\sqrt{x-2}$$

iv.

$$f(x) = e^{3x^2}$$

v.

$$f(x) = 4^{3x}$$

vi.

$$f(x) = \ln(2x^2 + 1)$$

vii.

$$f(x) = \log_2(5x + 1)$$

viii.

$$f(x) = \cos(2x)$$

ix.

$$f(x) = \sin(x^2 + 2)$$

x.

$$f(x) = \tan(2x + 1)$$